ConvertAudio2-Stereo.ps1 (v1.4.0)

> Downmixes any 5.1, 7.1, or Atmos track into a high-quality EAC3 or Opus stereo track.

- **PreTrack-QRS** — selects the best source using a quality ranking and a clear tie-break order (rank → language → bitrate → index) before any encoding decision is made.
- **Safe Peak Normalizer (SPN)** — scans first, applies loudnorm only when the source truly needs it, preserving dynamics on clean tracks.
- **Multichannel Quality Ladder** — ranks formats from TrueHD to AAC
- **Tie-Break Logic** — resolves equal scores using language, bitrate, and index
- **Malformed-layout correction** — fixes invalid 7-channel layouts
- **ITU-R BS.775 pan matrices** — accurate 5.1/7.1 > stereo downmixing.

PowerShell 7.6 - [Click here to Download PS 7.6 (Windows x64 .msi)]
FFmpeg 8.1 - [https://ffmpeg.org/download.html]

Quick Start (Windows 11)

1. Place `ConvertAudio2-Stereo.ps1`, `ffmpeg.exe`, `ffprobe.exe` in the same folder.
2. Run the script:

> pwsh -ExecutionPolicy Bypass -File .\ConvertAudio2-Stereo.ps1 ".\YourMovie.mkv"

The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, `ffmpeg`, and `ffprobe` in the same folder.
2. Make the binaries executable (first time only):

> chmod +x ffmpeg ffprobe

3. Run:

> pwsh -File ./ConvertAudio2-Stereo.ps1 "./YourMovie.mkv"

1. Stereo Bitrate Selection

You can switch codecs at runtime by passing -StereoCodec Opus or EAC3 on the command line. Runtime switching is temporary, while changes within the script are permanent.

```
pwsh -ExecutionPolicy Bypass -File .\ConvertAudio2-Stereo.ps1
".\YourMovie.mkv" -StereoCodec Opus
or
pwsh -ExecutionPolicy Bypass -File .\ConvertAudio2-Stereo.ps1
".\YourMovie.mkv" -StereoCodec EAC3
```

Remember to save file after changing Codecs or Bit-rates.

To change audio codec type, within the script. (line 25-27)

```
# Default stereo codec used when -StereoCodec is not provided.
# You can change this to "Opus" or "EAC3"
$DefaultStereoCodec = "EAC3"  << Change Codec from "EAC3" to "OPUS"
```

To change audio codec bit-rates, within the script (line 38-40)

```
# You can safely adjust these numbers, within "...k"
# EAC3: 224, 256, 320, 384, 448, 512
# Opus: 192, 224, 256, 320, 384, 448
"EAC3" = "384k"  << Change your EAC3 Bitrate
"Opus" = "320k"  << Change your OPUS Bitrate
```

EAC3 (384 = high-quality 2.0 / 448 = high-quality 5.1)
OPUS (192 = high-quality stereo / 320 = high-quality 5.1)

LFE fold-in is controlled by `$FoldLFE` in the Global Settings block (line 55).

```
When set to $true, the LFE channel is folded into the stereo mix at +0.5
gain inside the pan matrix.
When set to $false, the LFE channel is removed entirely.
[bool]$FoldLFE = $true  # $false discards LFE
```

Output filename: The program saves a new file in the same folder, adding a codec suffix like (_stereo_eac3.mkv) or (_stereo_opus.mkv). Your original file stays untouched.

2. Supported Codec Families

Family	Formats
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
PCM	pcm_s16le, pcm_s24le, pcm_f32le, pcm_f32be
FLAC	All FLAC layouts

3. FFmpeg Global Options

Flag	Purpose
`-threads 8`	Sets how many CPU threads the encoder uses. (4-16 user-adjustable)
`-analyzeduration 200M`	Deep probe for TrueHD/Atmos/AAC 7.1 layouts.
`-probesize 200M`	Prevents mis-detected channel layouts.
`-err_detect ignore_err`	Allows FFmpeg to continue through minor bitstream errors.
`-drc_scale 0`	Disables Dolby DRC for consistent loudness.
`-max_muxing_queue_size 14000`	Prevents muxer stalls on long files.
`-map 0:v? -c:v copy`	Always copy video losslessly. (no re-encode).
`-map_metadata 0 -map_chapters 0`	Preserve metadata and chapters.

4. ScanThrottle + Threadcount

Sets how many CPU threads the encoder uses. (line 47-49)

```
$ThreadCount     = 8    # User-adjustable (4-16).
```

Limits how many peak-scan tasks run at the same time.

```
$ScanThrottle = [Math]::Max(1, [int]([Environment]::ProcessorCount /
$ThreadCount))
Recommended values: 1, 2, 3, or 4, (replace 1,)
```

What ScanThrottle does

- Limits **parallel SPN scans** to avoid CPU overload
- Keeps the system responsive during peak detection
- Does **not** affect encoding speed

Parallel SPN scan — scans all tracks at the same time to detect peaks quickly.

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5. Safe Peak Normalizer (SPN)

SPN is a pre-filter safety system that checks each audio track for unsafe peaks before any downmix or encode occurs.

What SPN does

- Scans each non-Removed track using `volumedetect`
- Compares the detected peak against **−0.5 dBFS**
- Marks the track with **NeedsNormalization** when the peak is too high
- Inserts a light `loudnorm` step only when required
- **Clean sources** are Left untouched. Original loudness and dynamics are preserved.
- **Dirty sources** are Corrected safely without compression artifacts or "over-normalized" sound.
- Skips passthrough and commentary tracks entirely

5.5 SPN Behavior

Condition	Behavior
Peak is **greater than** −0.5 dBFS	Apply `loudnorm` before pan matrix
Peak is **less than or equal to** −0.5 dBFS	No normalization
Passthrough track	SPN skipped
Commentary track	SPN skipped

5.7 SPN Thresholds

To adjust **PeakThresholdDB**, change the value on line 62.

```
$PeakThresholdDB = -0.5 # SPN: loudnorm triggers when source peak exceeds this (dBFS)
```

- Valid range: **−1.5 dBFS to −0.3 dBFS** (recommended: −0.5 dBFS)
- Lower values (e.g. −1.5) normalize more tracks — more conservative.
- Higher values (e.g. −0.3) normalize fewer tracks — only the hottest sources.

Setting	Value
loudnorm target	−23 LUFS
Loudness range target	LRA=11 LU
True peak ceiling	−1.5 dBTP

6. QRS Integration

(QRS) stands for **Quality**, **Resolution**, and **Scoring**.
(SPN) stands for **Safe Peak Normalizer.**

- **SPN runs before QRS**, scanning all non-Removed tracks so peak and bitrate data are available for evaluation.
- SPN provides **peak**, **bitrate**, and **NeedsNormalization** flags to the QRS Quality, Resolution, and Scoring stages.
- **NeedsNormalization** is set from SPN peak data and controls whether the FFmpeg chain applies loudness correction.

6.5 PreTrack-QRS

PreTrack-QRS decides which single audio track becomes the stereo source.
It evaluates all candidates before any downmix or encode occurs.

How PreTrack-QRS works.

1. **Quality (Q)**

- Identifies all multichannel candidates and ranks them using the Quality Ladder
- (TrueHD > PCM/FLAC >> DTS-HD >>> EAC3 >>>> AC3 >>>>> AAC).
- Stereo codecs are ignored.

2. **Resolution of ties (R)**

- If two tracks share the same QualityRank, QRS applies a fixed chain:
- Language (eng > und >> others)
- Higher bitrate
- Lower stream index

3. **Scoring (S)**

- Applies structured checks (metadata, layout, profile) to finalize the winner.

After QRS completes:

- All non-winning tracks are marked **Removed**
- Only the single best track proceeds to encoding
- The script always produces exactly one stereo output

7. Multichannel Audio Ladder (Quality)

Higher = better. The Quality Ladder defines the fixed ranking used by QRS.

Score	Format / Channels	Score	Format / Channels
100	TrueHD Atmos 7.1/Atmos	80	DTS Core 5.1
99	TrueHD 7.1	75	EAC3 Atmos (JOC) 5.1/7.1
98	TrueHD 5.1	70	EAC3 7.1
95	PCM/FLAC 7.1	69	EAC3 5.1

94	PCM/FLAC 5.1	60	AC3 5.1
90	DTS-HD MA 7.1	50	AAC 7.1
89	DTS-HD MA 5.1	49	AAC 5.1
85	DTS-HRA 5.1/7.1		

8. ITU-R BS.775 Pan-Matrix Routing

The audio-engine uses ITU-R BS.775 pan matrices for all multichannel >> stereo downmixing.

5.1 > 2.0 Matrix

```
pan=stereo|
FL=FL+0.707*FC+0.707*BL+0.707*SL+0.5*BC+0.5*LFE|
FR=FR+0.707*FC+0.707*BR+0.707*SR+0.5*BC+0.5*LFE
```

7.1 > 2.0 Matrix

```
aformat=channel_layouts=7.1,pan=stereo|
FL=FL+0.707*FC+0.707*BL+0.5*SL+0.5*LFE|
FR=FR+0.707*FC+0.707*BR+0.5*SR+0.5*LFE
```

Why these matrices matter

- **0.707** = −3 dB fold for center/surround channels (ITU-R standard)
- **0.5** = LFE fold-in (optional via `$FoldLFE`)
- **aformat** ensures consistent 7.1 layout before Downmixing.
- **Limiter** prevents true-peak overshoots

> $alimiter = "alimiter=limit=0.948:attack=5:release=50:level=disabled:latency=1"

- **Latency**=1 allows the limiter a 1-sample look-ahead window to catch post-downmix peaks without adding audible delay.

9. EAC3 Conditional Passthrough

EAC3 2.0 is **only** passed through when:

1. The rule engine marks it as Passthrough
2. The track has exactly 2 channels
3. The user selected `-StereoCodec EAC3`

If any condition fails, the track is **re-encoded** to the chosen stereo codec.

10. Commentary Detection

Tracks are marked as Commentary when their metadata matches the commentary pattern:

Director, producer, writer, cast, behind, bonus, alt, interview, commentary,

Commentary tracks are excluded from QRS selection and cannot be chosen as the primary stereo source.

11. Malformed-Layout Guards

(Fixes illegal 7-channel layouts)

Some sources report invalid 7-channel layouts (AAC, EAC3, TrueHD, FLAC/PCM).

- When a malformed 7-channel layout is detected, the engine replaces it with a valid layout before QRS and downmixing.
- This prevents incorrect channel routing and ensures the pan matrix receives a consistent layout.

12. Fallback Logic

If no valid stereo candidate is found after QRS filtering, the engine falls back to the highest-ranked non-commentary track.
This ensures a consistent output when all stereo options are invalid or excluded.

13. Summary Report

At the end of processing, the audio-engine prints a detailed summary table.

This table includes:

Column	Description
Index	Internal track index (0-based).
Codec	Source codec (TrueHD, DTS, AAC, etc.).
Channels	Corrected channel count (after malformed-layout guard).
Action	Removed / Encode / Downmix / Passthrough.
Output	Final codec + bitrate + metadata tag.
Priority	Rule priority (0-100).
Rule	The rule that matched (or fallback).

Each row in the summary table is color-coded by action: **Color->Action**
Red->Removed, Green->Passthrough, Dark Cyan->Downmix, Blue->Encode

14. Non-Critical FFmpeg Warnings (Safe to Ignore)

FFmpeg may print warnings during probing or decoding.
These **do not** affect audio quality or sync.

Common warnings (safe to ignore)

Warning	Meaning	Impact
quant_step_size larger than huff_lsbs	TrueHD decoder diagnostic	Safe
Codec AVOption drc_scale … has not been used	DRC disabled intentionally	Safe
Subtitle: hdmv_pgs_subtitle unspecified size	PGS metadata missing	Safe
Assuming an incorrectly encoded 7.1 channel layout	AAC 7.1 variant	Safe (layout is corrected)
Could not find codec parameters for stream (pgssub)	PGS subtitle track has incomplete metadata	Safe
Consider increasing analyzeduration/probesize	Generic suggestion caused by incomplete PGS metadata	Safe
unspecified size (PGS)	Subtitle bitmap dimensions not declared	Safe

15. Audio-Script Rules That Must Not Change

a. Malformed-layout guards

AAC 7ch >> 6ch
EAC3 7ch >> 6ch
TrueHD 7ch >> 8ch
PCM/FLAC 7ch >> 8ch

b. Safe Peak Normalizer before PreTrack-QRS

Peak scanning completes before ranking begins.

c. Ranking table order

The Quality Ladder is evaluated top-down. The first match is selected.

d. Tie-break sequence

QualityRank >> Language >> Bitrate >> Index.

e. Pan-matrix operator

Pan matrices use the '=' operator to avoid auto-normalization.

f. Limiter position

The limiter is applied last in the audio chain.

g. EAC3 passthrough conditions

EAC3 2.0 is passed through only when StereoCodec=EAC3.

h. Commentary detection

Commentary detection applies only to 2-channel tracks.

i. Fallback rules must remain intact

Guarantees safe behavior on unknown codecs.

Disclaimer

- Always test on a few files before batch-processing your library.
- Keep backups of original media.
- Results may vary depending on source quality and hardware.